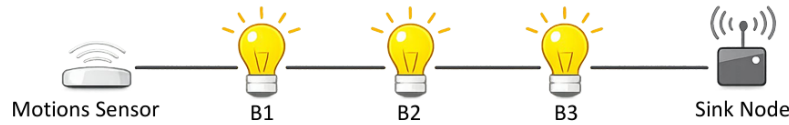


4 - Energy Trade-off in a WSN

The motion sensor has to transmit data to the SINK node. The transmission can occur directly or hop by hop thanks to the bulbs.



The total distance between the presence sensor and the sink is: $d = 40m$.

Nodes are equally spaced.

Each transmission of a $k = 2000 \text{ bit}$ packet follows the energy model:

- $E_{tx}(k, d) = kE_c + k\epsilon d^2 = 0.6 \text{ mJ} + 2000 \text{ bit} \cdot 1 \text{ nJ/bit/m}^2 \cdot (40\text{m})^2 = 3.8 \text{ mJ}$
- $E_{rx}(k) = kE_c = 2000 \text{ bit} \cdot 300 \text{ nJ/bit} = 0.6 \text{ mJ}$

where:

- $E_c = 300 \text{ nJ/bit}$
- $\epsilon = 1 \text{ nJ/bit/m}^2$

Assume ideal communication (no retransmissions, no idle energy).

Q1: Compute the total communication energy consumed by the network for one message in the case of: (a) direct transmission; (b) multi-hop transmission.

a) *Motion Sensor* $\rightarrow$ *Sink Node*

$$E_{DIR} = E_{tx}(k, d) + E_{rx}(k) = 3.8 \text{ mJ} + 0.6 \text{ mJ} = 4.4 \text{ mJ}$$

b) *Motion Sensor* $\rightarrow$ *B1* $\rightarrow$ *B2* $\rightarrow$ *B3* $\rightarrow$ *Sink Node*

$$E_{HOP} = [E_{tx, hop}(k, d_{hop}) + E_{rx}(k)] \cdot 4 = [0.8 \text{ mJ} + 0.6 \text{ mJ}] \cdot 4 = 5.6 \text{ mJ}$$

$$E_{tx, hop}(k, d_{hop}) = kE_c + k\epsilon d_{hop}^2 = 0.6 \text{ mJ} + 2000 \text{ bit} \cdot 1 \text{ nJ/bit/m}^2 \cdot \left(\frac{40\text{m}}{4}\right)^2 = 0.8 \text{ mJ}$$

Q2:

- Which strategy would you choose for long-term operation?
- Which performance metric are you optimizing when choosing the strategy for long-term operation?
- Provide a brief justification. Your explanation must clearly link the selected strategy to the chosen performance metric and be logically consistent with your numerical results.

The performance metric that we'd like to optimize is the lifetime of the motion sensor, then for long-term operation multi-hop strategy is preferable, because it can reduce the motion sensor energy consumption and makes the battery more durable.

Compared with the direct transmission, in our case, the sensor transmits over 10m instead of 40m and the energy consumption is 800uJ instead of 3800 uJ that corresponds to 86% of the total communication energy consumed.

In conclusion, although direct transmission minimizes the total network energy per message, multi-hop transmission is preferable for long-term operation because it significantly reduces the energy consumption of the only energy-constrained node, consequently extending the system lifetime.